Modern Statistical data analysis 52311, 2024-25
Final Project

To get full credit for a question, you need to solve it correctly and completely, and explain your answer clearly. For the computerized questions, you need to supply plots, explanations and your code. Once you receive the project packet, You have 5 days to submit the answers.

General remarks:

- You can either submit two files (PDF and ipynb) or a single python notebook file. **In both cases, the results have to be clearly explained:** What do we see in the plot/table? What are the conclusions one can obtain? Points will be deducted for plots without explanations.
- Filenames: Please name your files FirstnameLastname_FirstnameLastname.ipynb (i.e. ArielJaffe.ipynb),
- Unless stated otherwise, you can use python library functions of your choice. **However, you are expected to inspect the documentation** (input, parameters, output). Wrong use of functions (input parameters that make no sense etc.) will result in deduction of points.
- Please read carefully the LLM guideline section.

Supervised learning

In this section, you will analyze a dataset from the Israeli Bureau of Statistics - Economic literacy. The task is to predict, based on multiple features, if an individual has a debt. Explanation for the features are provided in the file *H20121242Codebook*

Remark. You can choose an alternative dataset from an online source or a research project. In this case, **before starting the project**, send me a short description with the following details: (i) where is the data from, (ii) what are the features and the labels, (iii) the data size

Please follow these steps.

1. *Preprocessing:* Details on the preprocessing steps - normalization, centering etc. are for you to determine. Please describe the preprocessing steps in the project report.

2. Compute and generate a plot of the Lasso path. Note that this is a classification and not a regression problem. (i.e. the standard lasso path function of sklearn is not applicable). Based on the path - what are the most important features for the prediction task?

3. In the next part, we would like to obtain insight into *groups of features* with large variance in the data.

   (a) Apply PCA to the matrix $X \in \mathbb{R}^{n \times p}$ to obtain the leading 4 (at least) principal components $v_i \in \mathbb{R}^p$ of the matrix X. These are the leading *eigenvectors* of the covariance matrix. What is the explained variance of each component?

   (b) For each of the computed principal components $v_i \in \mathbb{R}^p$, generate a plot of the absolute value of its elements $v_i(k)$ in descend ng order. For each vector - what are the subset of features whose corresponding elements in v_i have the largest absolute value?

   (c) Here, we would like to establish the relevance of each principal components to the labels y_i. Let $\alpha_i \in \mathbb{R}^n$ be a vector of coordinates that corresponds to the principal component v_i, such that $\alpha_i = Xv_i$. Compute the PR-AUC based on α_i and the vector of labels y. What can you say regarding the relevance of the principal components based on the results?

4. Separate the data into a training set that consists of approximately 2/3 of the observations, and a test set. Use the training set to train the following prediction models:

   - Logistic regression

   - K-nearest neighbors. Please specify the distance metric used (e.g., Euclidean, Manhattan).

   - Kernel Logistic or kernel SVM

Use cross validation (within the training set) with F1 score to determine the value of hyperparameters. Create a table summarizing both the training and test F1 scores for each algorithm.

5. In the next parts - we would like to obtain a computational measure of *diversity* between the three learners.

   (a) Generate, based on the training data, a matrix of 'errors' E of size $n \times 3$ (n is the size of the training data) for the three methods (Logistic, k-nn and Kernel logistic, n is the size of the training data). An entry E_{ij} is equal to 1 if a method j made a wrong prediction for observation i. Let $\tilde{E}$ equal the matrix E after centering and normalizing the columns of E such that,

   $$\sum_i \tilde{E}_{ij} = 0 \qquad \sum_i \tilde{E}_{ij}^2 = 1$$

   Compute and print the matrix $\tilde{E}^T \tilde{E}$. Based on this matrix, can we improve the results by combining the three learners?

   (b) Generate a *meta learner* - change to applying a new learner on the previous ones. combining the three learners with majority voting (i.e. the prediction of the meta learner is 1 if at least 2 learners predict 1). Has the testing results improved? Can you suggest a better way to combine the three learners? (no need to implement)

Unsupervised learning

In the next two sections you will analyze quarterly crime files datasets from Israel police crime files. Download the last file (2025). You will also need the 'cities.xls' file that contains various properties of different cities such as population size and coordinates. You can use the code from crime.ipynb for help in reading and merging data from the two sources.

1. Write a short paragraph about the dataset. Which features are saved for each act of crime?

2. The code from crime.ipynb reads the data and creates a dataframe with a table whose rows correspond to cities and columns correspond to types of crimes. Thus, each element specifies the number of crimes of type X commited city Y in 2025. Note that the grouping of crimes in the code is done according to the column 'StatisticGroupKod'.

3. Compute the Laplacian matrix of a graph whose nodes correspond to cities, and weights are computed based on similarity in crime profile. **Important:** explain exactly how you

computed the similarity between cities. Specifically, the similarity should be based on numbers of various crimes, but should also take into account the population size of each city, found in cities.xls. **Do not use any library function for computing the graph's weights**.

4. Compute the leading (smoothest) eigenvectors of the graph Laplacian. Generate a scatter plot where the points are located according to the (x,y) coordinates (see crime.ipynb) and colored by each of the (say 8) leading eigenvectors.

5. Compute spectral clustering of the cities - **do not use a spectral clustering library function**. Use the eigenvectors computed in the previous part and apply k-means (you may use a library function of k-means). Create a scatter plot where the points are colored according to the results of the clustering procedure.

6. Can you find features in 'cities.xls' that significantly correlate with the leading eigenvectors of the graph Laplacian matrix?

Remark The choice of kernel function is very significant to the outcome. For example, without normalization in the population size, the result will mainly highlight differences in the large cities.

Hypothesis testing/Multiple testing

For this question, observe the quarterly crime files in the Israel police crime files.

Here, you should come up with your own research question regarding the crime files. You can use multiple files from this source (e.g. 2023-2025). The research should include a **significant** element of multiple testing. You can also use (though it is not obligatory) other sources of data, such as the cities.xls file in Q2. In your submission, be sure to specify:

- What is the null and alternative hypothesis?
- What test did you apply? Why is this test adequate for your specific question and the data at hand?
- Is there a need to address multiple testing? What procedure is adequate for your question?
- Explain the results - are there any results that are statistically significant? What conclusion can you make based on your analysis?

Use of large language models (LLM) within the project

LLMs may be used to generate code for auxiliary or time-consuming components that are not central to the analytical work. For example, you may use them to create scripts for loading Excel files or improving the appearance of plots. However, you must not use LLMs to generate code for core analysis tasks (e.g., model fitting, statistical testing) or to generate any textual interpretation of the results. Please note: the quality of writing will not affect your grade, so there is no need to use LLMs for wording or language improvement. We reserve the right to check for compliance with these guidelines. Violations will result in disqualification of the project.

At the end of the submission file, add a short declaration regarding the use of LLM in the project. (i.e. I didn't use LLM at all or I used LLM for reading the input files etc.)